OLO ACADEMY · AI ENGINEERING PROGRAM (AIEP)

Capstone Technical Report · CallSense / Track E

NABRA AI

نبرة

Intelligent Call Analysis and Executive Reporting

An end-to-end platform that turns recorded customer-service calls into speaker-attributed transcripts, audited call analysis and an executive dashboard — automatic diarization, Whisper ASR, word-level alignment, DistilBERT role classification, Qwen2.5-3B analysis with an evidence-based resolution audit, and Streamlit reporting.

Team (3 students)	Lujain Al Mubarak · لجين آل مبارك Dalal Moataq · دلال معتق Ghada Al-Asiri · غاده عسيري
Programme	OLO Academy — AI Engineering Program (AIEP)
Project theme	CallSense / Track E — call-centre intelligence
Submission date	1 November 2026

Contents

Abstract 2
1. Introduction 2
2. Problem Statement 2
3. Project Objectives 3
4. Dataset 3
5. System Architecture 4
6. Data Preprocessing 4
7. Model Selection and Configuration 5
8. Training Procedure 5
9. Baseline Experiment 5
10. DistilBERT Results 6
10.1 Training 6
10.2 Test results 6
11. Whisper Evaluation 7
12. Call Analysis and Resolution Audit 8
12.1 Analysis output 8
12.2 Initial prediction vs audited status 8
12.3 Prompt engineering 8
13. Evaluation and Practical Tests 9
13.1 Historical speaker-role attribution evaluation 9
13.2 Evaluation framework for the latest version 9
13.3 Practical case studies (latest version) 9
13.4 Executive dashboard 10
14. Results Summary 10
14.1 Earlier experiment: TF-IDF domain classifier 10
15. Limitations 11
16. Responsible AI and Privacy 11
17. Requirements Traceability 12
18. Future Improvements 12
19. Conclusion 12
References 13

This report contains 7 figures and 9 tables.

Abstract

Managers of customer-service teams rarely have time to listen to full calls, and they often lack a simple view of how many calls were handled, what customers asked for and which issues remain open. **NABRA AI** is an end-to-end call-intelligence platform that turns recorded calls into speaker-attributed transcripts, structured call analyses and an executive dashboard. It was developed by a team of three in the OLO Academy AI Engineering Program (AIEP) under the CallSense / Track E theme.

The deployed pipeline combines **Pyannote Community-1** for automatic speaker diarization, **Whisper small.en** for speech recognition, word-level alignment of the transcript to speaker turns, a fine-tuned **DistilBERT** classifier with speaker-level consensus to label the **Agent** and the **Customer**, and **Qwen2.5-3B-Instruct** (FP16) for context-based domain identification, summarisation, customer-request extraction and resolution analysis. An evidence-based audit then checks the model's resolution prediction for each request. A **Streamlit** executive dashboard presents transcripts, domain and resolution statistics, speaker word counts, call durations and processing times, in Arabic or English and in light or dark mode.

On a 48-call research dataset (4,788 labelled segments, split by call 34 / 7 / 7), the fine-tuned DistilBERT reached **87.64% accuracy and 86.81% macro-F1** for segment-level speaker-role classification, compared with 82.01% and 80.85% for a TF-IDF + Logistic Regression baseline. Whisper small.en achieved a word error rate of **13.63%** on the seven test calls. In a separate historical evaluation on the same seven calls, the earlier speech pipeline attributed **98.36%** of transcribed segments to the correct speaker role; this figure concerns speaker-role attribution only. Domain identification and resolution auditing have been demonstrated on practical test calls but have not yet been validated on an independently labelled holdout set.

1. Introduction

Contact centres handle more calls than any supervisor can review. Each call can contain important requests or promised follow-ups, but these details are embedded in long conversations. NABRA AI applies speech recognition, speaker diarization and language models to convert each call into a structured, speaker-attributed summary, and aggregates the results in a dashboard so managers can see what happened across calls without listening to every recording.

As an illustration, reviewing 100 calls of 15 minutes each would take **25 hours** of listening.

2. Problem Statement

- **Manual review is slow.** Listening takes as long as the call itself, so supervisors can only check a small sample.
- **Calls are long.** Important details are mixed with greetings, waiting and repetition.
- **Follow-up is not visible.** Pending callbacks or actions are scattered through the conversation and are easy to miss.
- **Speakers are not identified.** A raw recording or transcript does not show who said what.
- **No clear overview.** Managers do not know how many calls were handled, and there is no clear dashboard with charts to support quick decisions.

3. Project Objectives

#	Objective	Status
O1	Prepare and validate the research call dataset	Implemented and evaluated
O2	Transcribe speech to text (Whisper small.en)	Implemented and evaluated
O3	Detect and separate speakers automatically (Pyannote Community-1)	Implemented
O4	Align transcribed words to speaker turns (word-level alignment)	Implemented
O5	Classify Agent vs Customer (DistilBERT)	Implemented and evaluated
O6	Reduce inconsistent role labels with speaker-level consensus	Implemented
O7	Identify the call domain from conversation context (Qwen)	Implemented; not yet validated
O8	Summarise the call and extract customer requests (Qwen)	Implemented; not yet validated
O9	Audit the resolution status of each request against transcript evidence	Implemented; not yet validated
O10	Executive dashboard with call statistics and charts (Streamlit)	Implemented

Table 1. *Project objectives and status. "Evaluated" refers to the quantitative experiments in Sections 9–11 and 13.1.*

4. Dataset

Model development used a **research dataset of 48 recorded customer-service calls** in Australian English from **13 business domains** (for example health, delivery, aviation, food, banking and insurance). Each call is annotated with speech segments, giving **4,788 segments** in total, each labelled **agent** (2,937) or **customer** (1,851).

We checked the data for missing text, invalid timestamps and duplicates and found none. We then split the data **by call**, so that no conversation appears in more than one set: **34 calls for training, 7 for validation and 7 for testing**. This 48-call dataset is used only for training and evaluating the models. It is separate from the calls uploaded to and processed by the dashboard.

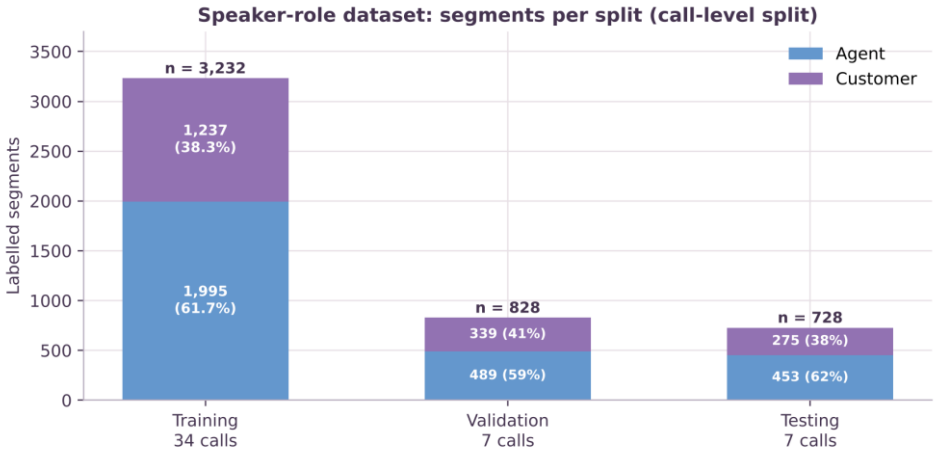


Figure 1. *Speech segments per split, by role (research dataset).*

5. System Architecture

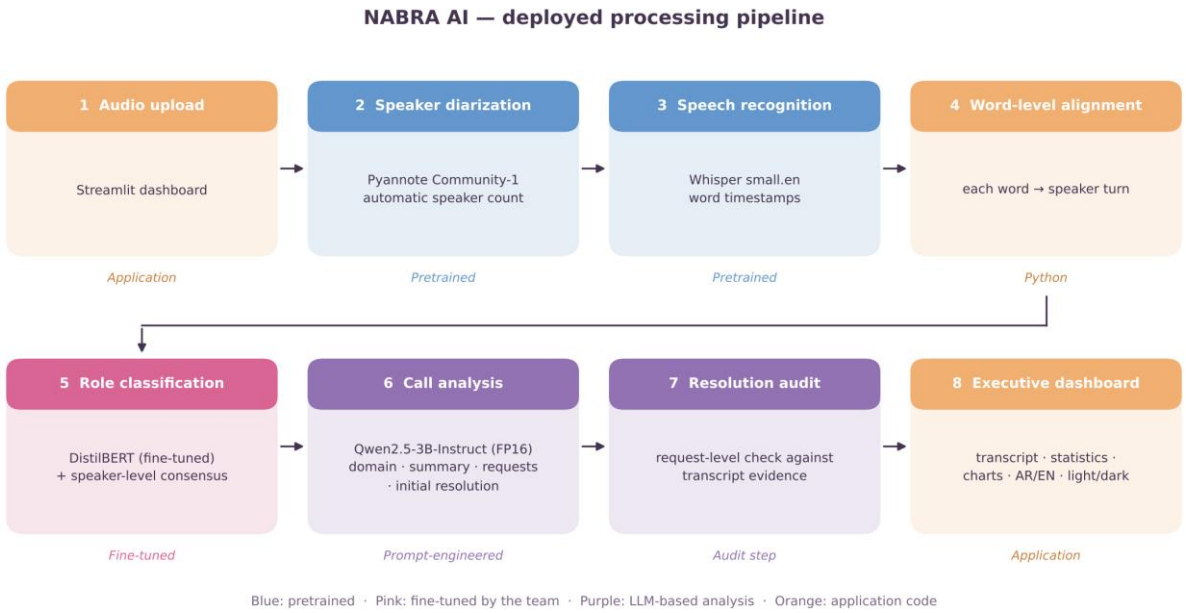


Figure 2. *NABRA AI deployed processing pipeline.*

A call passes through the following stages:

1. **Upload.** The user uploads an audio file in the Streamlit dashboard.
2. **Speaker diarization.** Pyannote Community-1 detects the speakers and their turns. The number of speakers is estimated automatically within a range of one to four, so the user does not need to enter it.
3. **Speech recognition.** Whisper small.en produces the transcript with word-level timestamps.
4. **Word-level alignment.** Each transcribed word is assigned to the speaker turn it falls in, which improves attribution when a Whisper segment spans a change of speaker.
5. **Role classification and consensus.** DistilBERT classifies each speaker's utterances, and a speaker-level consensus assigns one role to each speaker, reducing inconsistent labels.
6. **Call analysis.** Qwen2.5-3B-Instruct, running in FP16 on the GPU, identifies the call domain from context, summarises the call, extracts the customer's requests and gives an initial resolution prediction.
7. **Resolution audit.** An independent, evidence-based audit checks each request's predicted resolution against the transcript and produces the final audited status.
8. **Dashboard.** Results are stored and displayed with call-level details and aggregate statistics.

6. Data Preprocessing

- Loaded and validated the research audio (16 kHz WAV).
- Matched each recording to its annotation and extracted the labelled speech segments.
- Checked data quality: no missing text, invalid times or duplicates. 9.2% of segments overlap slightly with the previous speaker, which is normal in real calls.
- Split the data by call (34 / 7 / 7) with a fixed random seed.
- Prepared the text for DistilBERT: labels agent = 0 and customer = 1, maximum length 128 tokens.

7. Model Selection and Configuration

Model	Role in NABRA AI	Why it was chosen / configuration
Pyannote Community-1	Speaker diarization	Open-source diarization pipeline; automatic speaker count (1–4), exclusive speaker turns
Whisper small.en	Speech recognition	Strong pretrained English model; gave the best results when Whisper sizes were compared; provides word timestamps for alignment
DistilBERT (distilbert-base-uncased)	Agent / Customer classification	About 40% smaller and 60% faster than BERT-base with most of its accuracy; fine-tuned on the research dataset
Qwen2.5-3B-Instruct	Domain, summary, requests, initial resolution	Small open instruction model that follows structured prompts; runs in FP16 on the GPU
TF-IDF + Logistic Regression	Baseline	Simple reference to measure the gain from DistilBERT

Table 2. *Models and configuration.*

8. Training Procedure

Setting	Value
Base model	distilbert-base-uncased
Task	Binary classification: Agent / Customer
Data	3,232 training · 828 validation · 728 test segments
Max sequence length	128 tokens
Learning rate · batch size · weight decay	2e-5 · 16 · 0.01
Epochs	5
Model selection	Best epoch by validation macro-F1
Hardware	Google Colab, Tesla T4 GPU (training time 2 min 30 s)

Table 3. *DistilBERT training settings.*

All model results in Sections 9–11 come from this single training and evaluation run on the 48-call research dataset; figures and tables refer to the same run.

9. Baseline Experiment

As a reference, we trained a **TF-IDF + Logistic Regression** model on the same data. It reached **82.01% accuracy and 80.85% macro-F1** on the test set, showing that the task can be learned from wording alone and setting the level that DistilBERT must exceed.

Split	Accuracy	Macro-F1
Training	91.37%	91.06%
Validation	80.19%	79.30%
Testing	82.01%	80.85%

Table 4. *Baseline results (research dataset).*

10. DistilBERT Results

10.1 Training

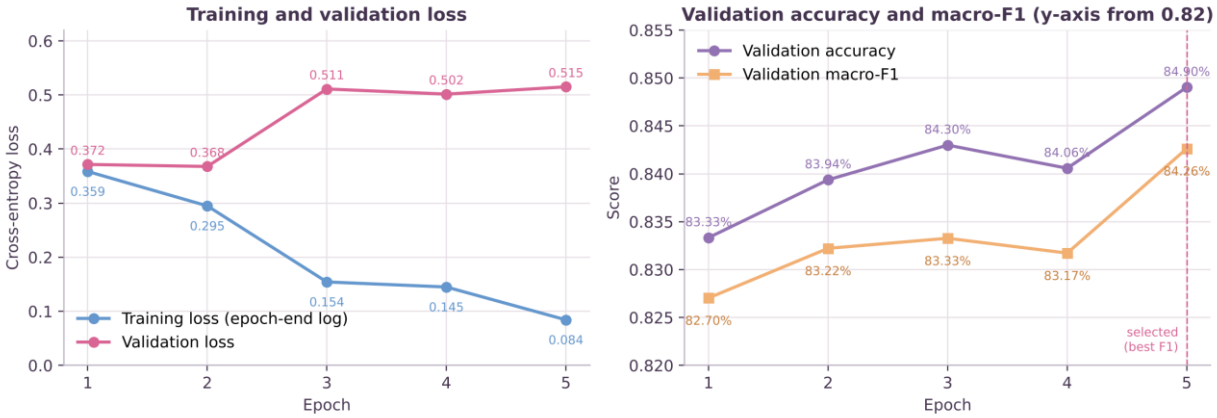


Figure 3. Training and validation loss (left) and validation accuracy and macro-F1 (right).

Training loss fell steadily, and validation macro-F1 reached **84.26%** at epoch 5, the selected model. Validation loss rose after epoch 2, indicating growing confidence on validation errors; the model was therefore selected by F1 rather than by loss, and its probability outputs should not be treated as calibrated.

10.2 Test results

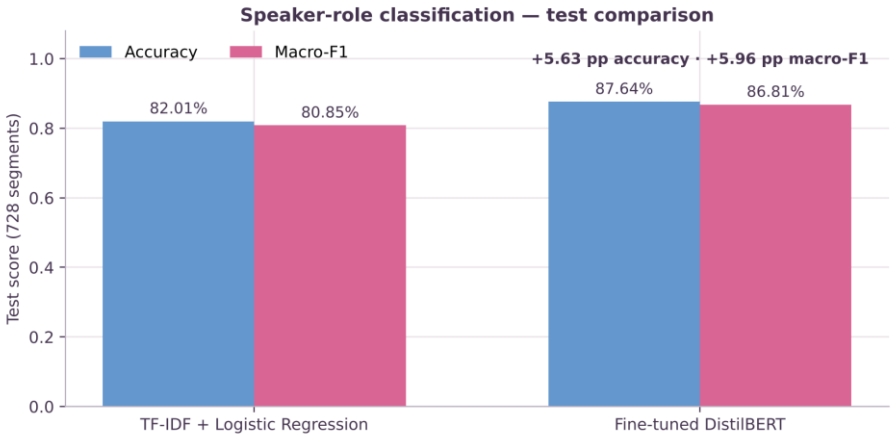


Figure 4. Baseline vs fine-tuned DistilBERT on the test set (728 segments).

Model	Accuracy	Macro-F1	Errors (of 728)
TF-IDF + Logistic Regression	82.01%	80.85%	131
Fine-tuned DistilBERT	87.64%	86.81%	90
Improvement	**+5.63 pp**	**+5.96 pp**	**−41**

Table 5. Segment-level speaker-role classification on the test set.

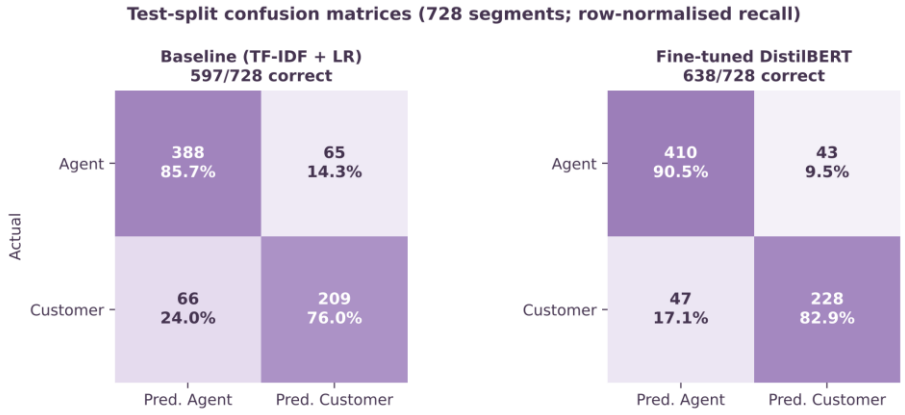


Figure 5. *Test confusion matrices.*

DistilBERT reduced errors by about a third; the largest gain was on the Customer class, where F1 rose from 0.76 to 0.84.

11. Whisper Evaluation

We first compared Whisper model sizes; **small.en gave the best results**, so it was selected. We then evaluated it on the 7 test calls against the reference transcripts. **Word error rate (WER)** is the share of words substituted, missed or added, and **character error rate (CER)** is the same for characters; lower is better for both.

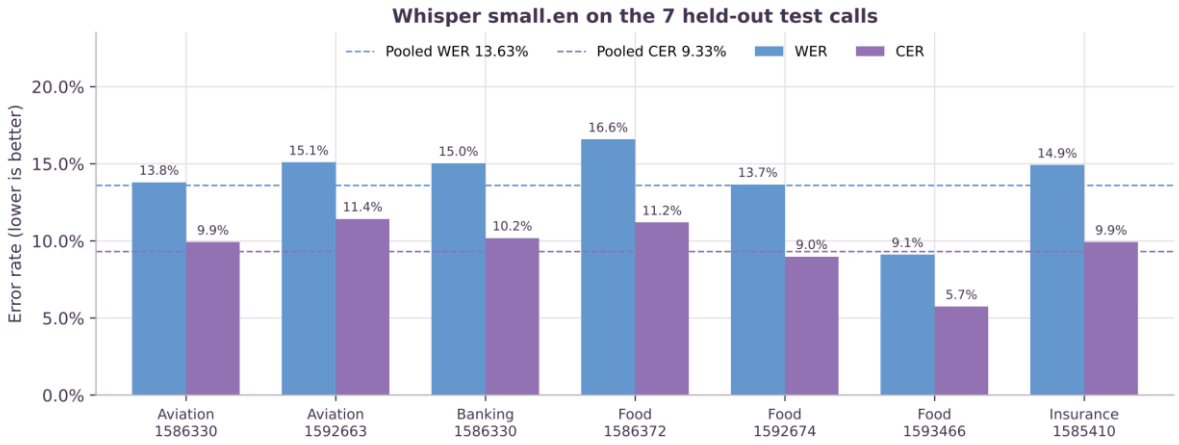


Figure 6. *Whisper small.en WER and CER per test call.*

Overall **WER was 13.63%** and **CER 9.33%**, with WER between 9.1% and 16.6% per call. Most errors involved filler words (such as “uh”) and company names.

12. Call Analysis and Resolution Audit

12.1 Analysis output

Qwen2.5-3B-Instruct reads the role-labelled transcript and produces a structured analysis. The audit step then adds the final status. For each call, the platform returns:

Field	Content
audio_file · total_segments · transcript[]	The call and its speaker-labelled transcript (start, end, speaker, role, text per line)
domain · subdomain	Business domain and sub-domain identified from the conversation context
summary	Short description of what happened in the call
issues[].request	Each customer request, extracted as a separate item with the transcript lines it comes from
issues[].model_status	Initial LLM prediction for the request
issues[].audit_outcome	Audit finding: completed, explicit_failure or not_confirmed
issues[].resolution_status	Audited status: resolved, unresolved or unclear
issues[].evidence · reason	Quoted transcript line (line id, role, quote) and the explanation
resolution_status · review_required	Overall audited status of the call and whether it needs human review
models	Model used at each stage

Table 6. *Main fields of the analysis report produced by the backend.*

12.2 Initial prediction vs audited status

The **initial resolution** is the language model's own judgement and may be optimistic, for example treating a promise as a completed action. The **audited resolution** is produced independently by a separate auditor prompt, which must quote the transcript line that supports its finding; the quote is accepted only if it appears word for word in the cited line. A request is marked **resolved** only when the quoted line shows a completed action, with no failure or tentative wording, and the request is not compound. It is marked **unresolved** only when the agent explicitly states that it cannot be done. In all other cases it is **unclear** and flagged for review; very long transcripts are sampled and also flagged. The dashboard reports the audited status.

Example: in the Event Catering call, the request “Get a quote for catering services” was first predicted as **unresolved** by the model. The audit changed it to **unclear**: the agent asked for event details, but no quote was given or refused during the call.

12.3 Prompt engineering

The prompts were developed iteratively: a fixed output structure, instructions to cover every request, reading events in order to find the final agreement, and testing on calls from different domains. The rules require the model to use only information present in the transcript and to treat a promise to follow up as not yet resolved.

13. Evaluation and Practical Tests

13.1 Historical speaker-role attribution evaluation

Before the latest pipeline changes, we ran the speech pipeline (Whisper, Pyannote with two speakers specified and DistilBERT with per-speaker averaging) directly from audio on the **7 test calls of the research dataset**. **98.36%** of the 916 transcribed segments were attributed to the correct speaker role (macro-F1 98.26%).

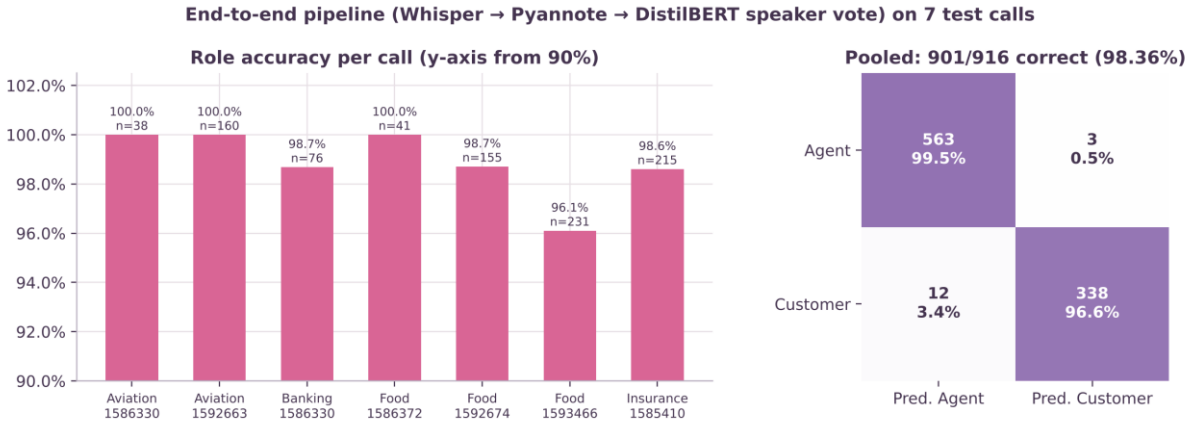


Figure 7. *Historical evaluation: role-attribution accuracy per call and overall confusion matrix.*

This result measures **speaker-role attribution only**, on these seven calls. It is not an overall accuracy of NABRA AI, and it does not measure domain identification, summarisation, request extraction or resolution detection. The deployed pipeline adds automatic speaker counting and word-level alignment, and has not been re-scored on this benchmark.

13.2 Evaluation framework for the latest version

A quality check on the stored reports confirmed that every evidence quote matches its transcript line. An accuracy-evaluation step compares domain and resolution outputs with reference labels and keeps development calls separate from an independent holdout set. At the time of writing, the reference labels have not been completed and the holdout set is empty, so **domain and resolution accuracy are not yet reported**.

13.3 Practical case studies (latest version)

The latest version was tested on several calls. These are observed examples, not a statistically representative benchmark.

Call	Observed output	Audited resolution	Processing time
Event Catering (Brunch Catering)	2 speakers detected; customer 141 words, agent 73 words; 0 speaker-role conflicts after consensus (2 segments corrected)	Unclear	≈ 35 s
Healthcare	3 customer requests extracted	Unclear	≈ 3.4 min
Real Estate	Real-estate domain identified	Unclear	≈ 1.3 min
Agriculture / Government Subsidies	Subsidy-related call identified	Unclear	≈ 1.6 min

Table 7. *Observed results from the latest tests.*

In all four cases the pipeline produced a complete analysis, and the audited status was **unclear**. Without labelled ground truth for these calls, it cannot yet be determined whether this reflects the calls themselves or a conservative audit. Processing time varied between about half a minute and a few minutes per call.

13.4 Executive dashboard

The Streamlit dashboard brings the pipeline together for managers:

- **Upload and analysis:** upload a call and run the full pipeline.

- **Call explorer:** review each processed call with its speaker-labelled transcript, domain, requests and resolution.
- **Statistics:** speaker word counts, call duration and processing time for each call.
- **Charts:** distribution of domains and resolution statuses across processed calls.
- **Interface:** Arabic / English switching and light / dark appearance.

> **Screenshot placeholder — Upload and analysis**
> Upload screen and the analysis view for one call.
> [Insert actual Streamlit screenshot here before submission; remove personal data.]

> **Screenshot placeholder — Call explorer**
> Speaker-labelled transcript with domain, requests and audited resolution.
> [Insert actual Streamlit screenshot here before submission; remove personal data.]

> **Screenshot placeholder — Dashboard charts**
> Domain and resolution distributions with call statistics.
> [Insert actual Streamlit screenshot here before submission; remove personal data.]

14. Results Summary

Component	Task	Result	Scope
TF-IDF + Logistic Regression	Speaker role (segments)	Acc 82.01% · F1 80.85%	Research test set
DistilBERT	Speaker role (segments)	Acc 87.64% · F1 86.81%	Research test set
Whisper small.en	Speech recognition	WER 13.63% · CER 9.33%	7 research test calls
Earlier speech pipeline	Speaker-role attribution	Acc 98.36% · F1 98.26%	Historical, 7 research test calls
Qwen + audit	Domain, requests, resolution	Case studies only	Not yet validated

Table 8. *Main results and their scope.*

14.1 Earlier experiment: TF-IDF domain classifier

Before adopting context-based domain identification with Qwen, we tested a TF-IDF + Logistic Regression domain classifier on the research dataset. With only 2–6 calls per domain, some domains had no training calls (for example Food), and the classifier reached 42.9% on validation and 14.3% on test. The deployed version therefore identifies the domain from the conversation context with the language model.

15. Limitations

- **Transcription errors:** Whisper makes errors on names and filler words, which can propagate to the analysis.
- **Diarization and role errors:** speaker turns or roles can be wrong, especially with overlapping speech or more than two speakers.
- **Incomplete request extraction:** the language model may miss or merge customer requests.

- **Hallucination risk:** the language model may state information that is not in the transcript; the audit and the visible transcript mitigate but do not remove this risk.
- **Insufficient evidence:** when the transcript does not clearly confirm an outcome, the audited status is unclear, as in all four case studies.
- **No independent end-to-end validation:** domain identification, request extraction and resolution accuracy have not yet been measured on an independently labelled holdout set; quantitative results cover speaker roles and transcription only.
- **Small evaluation samples:** seven test calls for the quantitative evaluations and four practical case studies.

16. Responsible AI and Privacy

- Calls contain personal information such as names and dates of birth; they should be protected and handled under Saudi Arabia's Personal Data Protection Law (PDPL).
- Only authorised calls should be used in demonstrations and screenshots.
- Automatic masking of personal data is recommended before storage or display; it is not part of the current version.
- Reports should be labelled as AI-generated and shown alongside the transcript for checking.
- Reports should not be used on their own to evaluate individual agents.

17. Requirements Traceability

Requirement	NABRA AI	Status
Speech recognition	Whisper small.en (selected after size comparison)	Implemented and tested
Speaker diarization	Pyannote Community-1, automatic speaker count	Implemented
Speaker attribution of text	Word-level alignment	Implemented
Speaker-role identification	DistilBERT + speaker-level consensus	Implemented and tested
Baseline vs advanced model	TF-IDF + LR vs DistilBERT	Implemented and tested
Domain identification	Context-based, Qwen2.5-3B	Implemented; not yet validated
Call summarisation and request extraction	Qwen2.5-3B (FP16)	Implemented; not yet validated
Resolution and follow-up status	Initial LLM prediction + evidence-based audit	Implemented; not yet validated
Executive dashboard	Streamlit: statistics, charts, AR/EN, light/dark	Implemented
Processing-time tracking	Shown per call in the dashboard	Implemented

Table 9. *Requirements and implementation status.*

18. Future Improvements

1. **Independent validation:** label a holdout set of calls for domain, requests and resolution, and measure end-to-end accuracy.
2. **Re-evaluate the deployed speech pipeline** (automatic speaker count, word-level alignment) on the research test calls.

3. **Privacy safeguards:** automatic masking of personal data, access control and data-retention rules.
4. **Deployment hardening:** reliability, monitoring and faster processing for long calls.
5. **More data** per domain and per call type.

19. Conclusion

NABRA AI has grown from a speaker-role classifier into a complete call-intelligence platform: automatic diarization, speech recognition, word-level alignment, role classification with speaker consensus, LLM-based domain, request and resolution analysis with an evidence-based audit, and an executive dashboard. Its measured components perform well on the research dataset: DistilBERT identifies agents and customers with **87.64%** accuracy, about six points above the baseline, and Whisper small.en transcribes with a **13.63%** word error rate. Practical tests show the full platform producing complete analyses within seconds to a few minutes per call. The next step is an independently labelled evaluation of domain, request and resolution accuracy before the platform is relied on for operational decisions.

References

[1] Bredin, H., et al. (2020). pyannote.audio: Neural building blocks for speaker diarization. *Proc. IEEE ICASSP 2020*, 7124–7128.

[2] pyannote (2025). speaker-diarization-community-1 [model card]. Hugging Face.

[3] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2023). Robust speech recognition via large-scale weak supervision. *Proc. ICML 2023*, PMLR 202, 28492–28518.

[4] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. *arXiv:1910.01108*.

[5] Qwen Team (2024). Qwen2.5 technical report. *arXiv:2412.15115*.

[6] Wolf, T., et al. (2020). Transformers: State-of-the-art natural language processing. *Proc. EMNLP 2020: System Demonstrations*, 38–45.

[7] Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *JMLR*, 12, 2825–2830.

[8] Streamlit Inc. Streamlit: an open-source app framework for machine learning and data science. https://streamlit.io

[9] Saudi Data & AI Authority (SDAIA). Personal Data Protection Law and its Implementing Regulations.

[10] AppTek. Call-centre speech data, Australian English (research dataset).